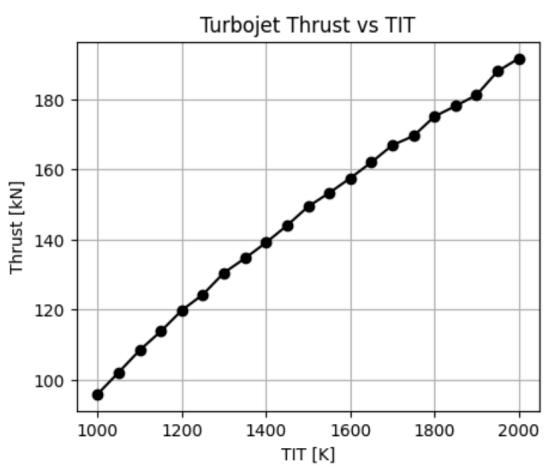
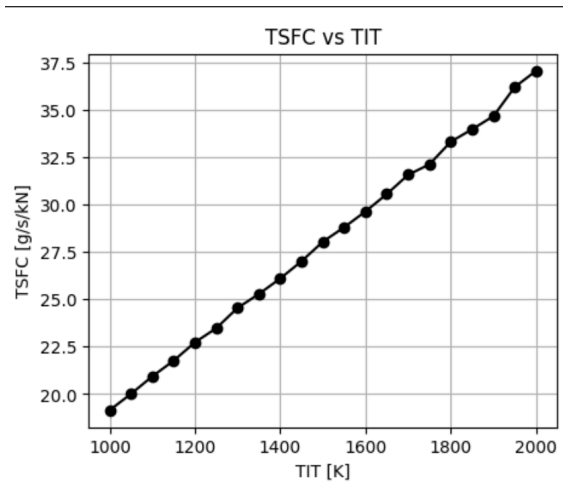
Project
Aerospace Propulsion
Course # 16805
David Reyna
Fall 2025

Tumansky RD-3M-500

Task 1

I decided to go with the Tumansky RD-3M-500 turbo jet engine. This gave me an airflow rate of 163.3 and a compression ratio of 6.4. I was not able to find exact values for some parameters so I chose the following estimates; for compressor efficiency I chose 0.87, combustor efficiency of 0.98, combustor pressure loss of 0.05, turbine efficiency of 0.91, and finally jet pipe loss of 0.03. As a base, it uses methane gas as its fuel source. I also used sea level conditions (meaning Po = 101.3 kPa, and To is 288.2 kelvin). There is also no inlet velocity (Vo=0). This gave me the theoretical ideal values of the ramjet thrust and its thrust specific fuel consumption ratio. The results showed an average of 147.3 kN of thrust and a TSFC of 27.7 g/s/kN, as well as a peak of 191.66 kN of thrust and a TSFC of 37.04 g/s/kN at 2000 kelvin. These outputs are relatively consistent with the behavior of turbojet engines operating within similar pressure ratios and temperature ranges. These values confirm that the chosen parameters provide a realistic representation of engine performance under idealized sea-level conditions.





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Task 2

After running the Colab simulation and confirming that my input values were feasible and realistic, we can observe consistent results. When comparing the data from the engine's specification sheet to my own simulated ideal engine results, a clear discrepancy appears between the two sets of values. At the same TSFC, the thrust outputs are not identical. According to the spec sheet, at a TSFC of 28.3 g/kN/s, the engine should produce a thrust of 93.2 kN. However, in my simulation, when the TSFC was 28.8 g/kN/s, the thrust reached 153 kN. This represents a difference of roughly 40%, which is significant.

There are likely two main reasons for this discrepancy. The first is the limited availability of public information about this engine. During research, I had to make informed estimates for several key parameters—specifically, compressor efficiency, combustor efficiency, combustor pressure loss, turbine efficiency, and jetpipe loss. The difference between these assumed values and the actual engine-specific data could easily account for much of the variation. The second possible reason is the experimental conditions under which the engine was originally tested. If the testing environment was not ideal, it may have caused deviations from theoretical performance, explaining part of the error but likely not all of it.

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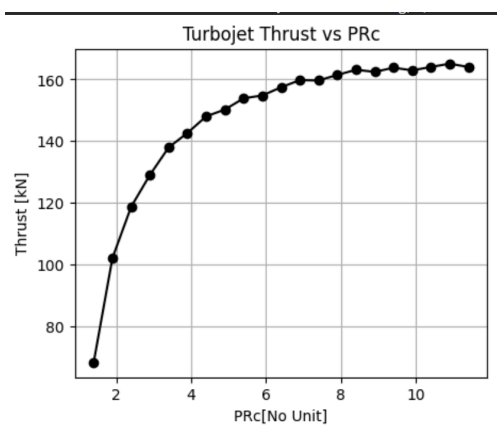
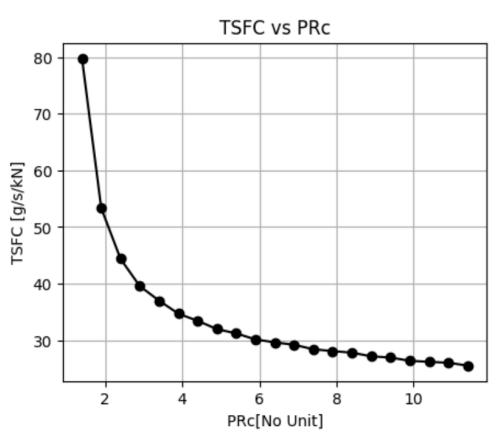
Task 3

Turbine Inlet Temperature (TIT) is not the only factor that affects thrust and thrust specific fuel consumption (TSFC). The compressor pressure ratio (PRc) also has a major influence, sometimes even more than TIT itself. For this analysis, I simulated how different compression ratios impact thrust and TSFC at a set TIT of 1600K. I used a range of PRc values centered around the engine's specific pressure ratio of 6.4 as the midpoint, varying it by about ±5 points.

After running the simulation, the trend became clear. As PRc increases, the thrust also increases. This happens because a higher PRc raises the total pressure before combustion. With more pressure available, the gases can expand more forcefully through the turbine and nozzle, resulting in a higher exhaust velocity and greater thrust output. The increase in thrust is significant up to the design PRc of 6.4, after which it continues to rise but at a slower rate.

As for the TSFC, this value decreases as PRc increases. A higher PRc value improves both thermal and combustion efficiency, meaning less fuel is required to reach the same turbine inlet temperature. With less fuel consumed per unit of thrust, the TSFC decreases, resulting in a more efficient overall system.





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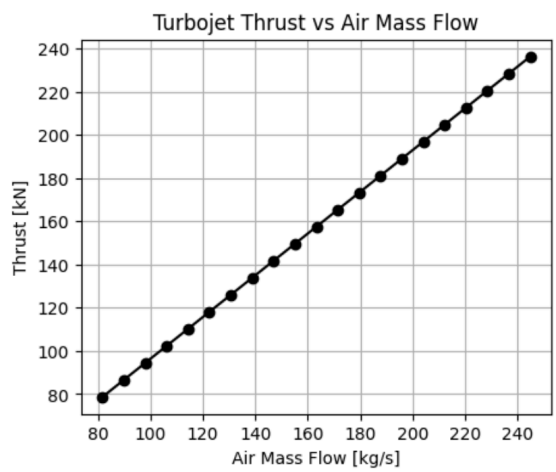
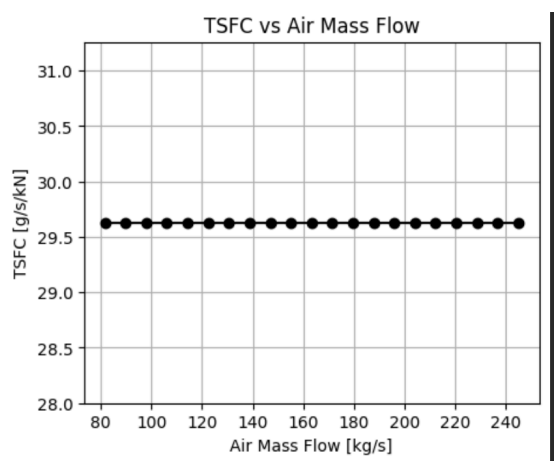
Task 4

A critical factor in the engine cycle is the mass airflow entering the engine. Thrust is directly proportional to the incoming airflow, as expressed by $F_n = M_{air}(\Delta V) + \Delta P A_{exit}$. As airflow increases, more fuel can be added for higher thrust, but the fuel mass flow does not increase proportionally, so TSFC should decrease and system efficiency should improve. For this analysis, I used the engine's specification of 163.3 kg/s and varied it ±50%, keeping the turbine inlet temperature fixed at 1600 K for consistency with other analyses.

After reviewing the outputs, the results appear unrealistic. The thrust is calculated mainly as F_n=(mdotait+mdotfuel)*V6, and since T5 and T6 barely change, the exhaust velocity remains nearly constant, leading to a roughly linear increase in thrust with airflow. TSFC also remains nearly constant because mdotfuel is set as a fixed fraction of mdotair / AFR, so the ratio mdotfuel/Fn does not change significantly. The main issue is that the simulation does not recalculate mdotfuel to achieve the target TIT for each airflow, failing to reflect thermodynamic constraints accurately. I attempted to correct this using more realistic models and external references, but a fully realistic implementation was not achieved.

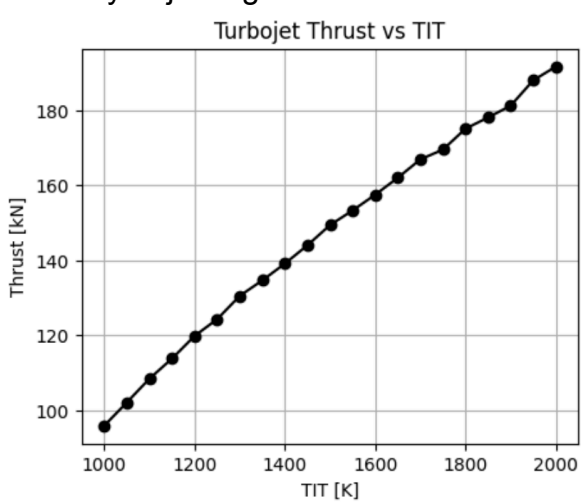


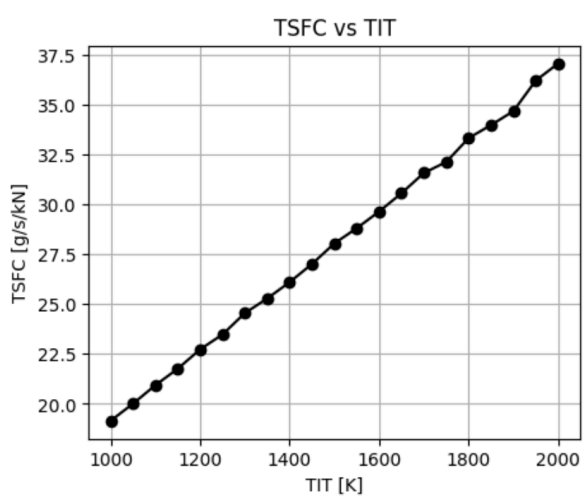


Task 5

Another parameter that significantly affects both thrust and TSFC is the turbine inlet temperature (TIT), which represents the temperature of the gases after combustion and before entering the turbine. As the TIT increases, the exhaust gases contain more energy in the form of enthalpy, which can then be expanded through the turbine and converted into useful kinetic energy. This higher energy content translates to a higher exhaust velocity, and therefore greater thrust output. In theory, this means that increasing TIT directly improves the engine's thrust-producing capability. Additionally, a higher TIT improves the thermal efficiency of the engine cycle because it increases the temperature difference between the combustor and the compressor inlet. This larger temperature ratio allows a greater portion of the fuel's chemical energy to be converted into mechanical work, improving overall efficiency and reducing the thrust specific fuel consumption (TSFC). In other words, for a given amount of fuel, the engine produces more thrust, meaning it operates more efficiently.

However, in this mode, like the one for the variation in airflows, the results deviates from the expected physical behavior. While thrust rises with TIT as expected, the TSFC also increases instead of decreasing. This suggests an issue with how the code handles fuel flow or temperature equilibrium, as the simulation may not be correctly adjusting the fuel-air ratio to maintain the targeted TIT for each iteration.





Final Code

```
import numpy as np
import matplotlib.pyplot as plt
import cantera as ct

# --- Design Parameters (Set your own values here) ---
T0 = 288.2        # Ambient temperature, K
P0 = ct.one_atm    # Ambient pressure, Pa
V0 = 0             # Vehicle velocity, m/s
mdotair = 163.3       # kg/s, example
PRc = 6.4           # Compressor pressure ratio
Effc = 0.87        # Compressor efficiency
Effcomb = 0.98     # Combustor efficiency
dPcomb = 0.05      # Combustor pressure loss (fraction)
Efft = 0.91        # Turbine efficiency
dPjp = 0.03        # Jetpipe pressure loss (fraction)
#TIT=1600
# --- Initialize storage ---
F_map = []
TSFC_map = []

# --- Loop over Turbine Inlet Temperatures (TIT) ---
for TIT in np.linspace(1000, 2000, 21):
#for PRc in np.linspace(1.4, 11.4, 21):
#for mdotair in np.linespace(81.1,245.5,21)
    gas = ct.Solution('gri30.yaml')
    Yair = np.zeros(gas.n_species)
    Yair[gas.species_index('O2')] = 0.233
    Yair[gas.species_index('N2')] = 1 - 0.233

    # --- State 1: Compressor Inlet ---
    gas.TPY = T0, P0, Yair
    cp0 = gas.cp_mass
    gamma0 = gas.cp_mass / gas.cv_mass
    Tt1 = T0 + 0.5 * V0**2 / cp0
    Pt1 = P0 * (Tt1 / T0)**(gamma0 / (gamma0 - 1))

    # --- State 2: Compressor Exit ---
    gas.TPY = Tt1, Pt1, Yair
    cp1 = gas.cp_mass
    gamma1 = gas.cp_mass / gas.cv_mass
```

```
    Pt2 = PRc * Pt1
    Tt2s = Tt1 * PRc**((gamma1 - 1) / gamma1)
    Tt2 = Tt1 + (Tt2s - Tt1) / Effc
    wdotc = cp1 * (Tt2 - Tt1)

    # --- State 3: Combustor Exit ---
    Yfuel = np.zeros(gas.n_species)
    Yfuel[gas.species_index('CH4')] = 1
    Tt3 = Tt2
    AFR = 1000

    while Tt3 < TIT:
        AFR -= 1
        mdotfuel = mdotair / AFR
        gas.TPY = Tt2, Pt2, Yfuel + Yair * AFR
        gas.equilibrate('HP')
        Tt3s = gas.T
        Tt3 = Tt2 + Effcomb * (Tt3s - Tt2)

    Pt3 = gas.P * (1 - dPcomb)
    Yexh3 = gas.Y
    cp2 = gas.cp_mass
    qdotcomb = cp2 * (Tt3 - Tt2)

    # --- State 4: Turbine Exit ---
    gas.TPY = Tt3, Pt3, Yexh3
    cp3 = gas.cp_mass
    gamma3 = gas.cp_mass / gas.cv_mass
    Tt4 = Tt3 - AFR * wdotc / ((AFR + 1) * cp3)
    Tt4s = Tt3 - (Tt3 - Tt4) / Efft
    Pt4 = Pt3 * (Tt4s / Tt3)**(gamma3 / (gamma3 - 1))

    # --- State 5: Jetpipe Exit ---
    Tt5 = Tt4
    Pt5 = Pt4 * (1 - dPjp)

    # --- State 6: Nozzle Exit ---
    gas.TPY = Tt5, Pt5, Yexh3
    cp5 = gas.cp_mass
    gamma5 = gas.cp_mass / gas.cv_mass
    R5 = ct.gas_constant / gas.mean_molecular_weight
    PRcrit = ((gamma5 + 1) / 2)**(gamma5 / (gamma5 - 1))
```

```
    if Pt5 / P0 < PRcrit:
        # Not choked
        P6 = P0
        T6 = Tt5 * (P6 / Pt5)**((gamma5 - 1) / gamma5)
        V6 = np.sqrt(2 * cp5 * (Tt5 - T6))
        A6 = (mdotfuel + mdotair) * R5 * T6 / (P6 * V6)
    else:
        # Choked flow
        P6 = Pt5 / PRcrit
        T6 = Tt5 * (P6 / Pt5)**((gamma5 - 1) / gamma5)
        V6 = np.sqrt(2 * cp5 * (Tt5 - T6))
        A6 = (mdotfuel + mdotair) * R5 * T6 / (P6 * V6)

    # --- Performance ---
    F = ((mdotair + mdotfuel) * V6 - mdotair * V0 + A6 * (P6 - P0)) / 1e3  # Thrust
[kN]
    TSFC = mdotfuel / F * 1e3  # [g/s/kN]

    if np.isnan(F):
        print("Engine not feasible for TIT =", TIT)
    else:
        print(f"TIT = {TIT:.0f} K → Thrust = {F:.2f} kN, TSFC = {TSFC:.2f} g/s/kN")

    F_map.append(F)
    TSFC_map.append(TSFC)

# --- Plot results ---
plt.figure(figsize=(5,4))
plt.plot(np.linspace(1000,2000,21), F_map, 'o-k')
#plt.plot(np.linspace(1.4, 11.4, 21), F_map, 'o-k')
#plt.plot(np.linspace(81.1, 245.5, 21), F_map, 'o-k')
plt.xlabel('TIT [K]')
#plt.xlabel('PRc[No Unit]')
#plt.xlabel('Airflow [kg/s]')
plt.ylabel('Thrust [kN]')
plt.title('Turbojet Thrust vs TIT')
#plt.title('Turbojet Thrust vs PRc')
#plt.title('Turbojet Thrust vs Airflow')
plt.grid(True)
plt.show()
```

```
plt.figure(figsize=(5,4))
plt.plot(np.linspace(1000,2000,21), TSFC_map, 'o-k')
#plt.plot(np.linspace(1.4, 11.4, 21), TSFC_map, 'o-k')
#plt.plot(np.linspace(81.1, 245.5, 21), F_map, 'o-k')
plt.xlabel('TIT [K]')
#plt.xlabel('PRc[No Unit]')
#plt.xlabel('Airflow [kg/s]')
plt.ylabel('TSFC [g/s/kN]')
plt.title('TSFC vs TIT')
#plt.title('TSFC vs PRc')
#plt.title('TSFC vs Airflow')
plt.grid(True)
plt.show()
```